CS 372 Project Proposal: A Council of Agents for Causal Reasoning and Portfolio Construction in the Semiconductor Value Chain

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Problem Statement and Motivation

Fundamental analysis of equities in the semiconductor industry requires synthesizing information from disparate, specialized domains, from geopolitics and monetary policy to fabrication technology and commodity markets, and translating these insights into specific, quantifiable adjustments to financial model inputs. This project addresses this challenge by designing a Council of Agents, a multi-agent system where five domain-specialized agents monitor their respective information sources and collaborate to construct causal graphs of events and their downstream effects. A sixth agent, the Portfolio Manager, synthesizes the council's insights, translates them into adjustments to consensus financial projections (revenue growth, margins, capex) for five semiconductor companies, feeds these into DCF models to generate intrinsic valuations, and constructs a long portfolio by buying stocks when intrinsic value exceeds market price. Portfolio performance is benchmarked against the S&P 500.

Our central research questions are: Can a council of specialized domain agents beat the S&P 500 by constructing accurate causal graphs and generating superior investment returns? How does multi-agent collaboration improve reasoning quality and financial performance? What are the characteristic failure modes of this collaborative reasoning and portfolio construction process?

Related Work and Background

Recent work in multi-agent systems has shown that collaboration between multiple LLM agents can improve performance on complex tasks. Du et al. (2023) demonstrated that multi-agent debate improves factuality and reasoning in language models, while this course's AGI framework emphasizes collaborative intelligence as a key component of advanced AI systems. This project applies these concepts to fundamental financial analysis, a domain naturally suited for multi-agent reasoning due to its reliance on synthesizing unstructured information from multiple specialized fields. The architecture mirrors professional investment teams, where domain experts (macro strategist, technology analyst, commodities trader) provide inputs to a portfolio manager who makes final allocation decisions. We evaluate the system on five companies representing the semiconductor and AI value chain: Cadence (EDA), Coreweave (Neocloud), NVIDIA (GPU Design), TSMC (Foundry), and ASML (Equipment). The evaluation is grounded in 5-10 major historical events (e.g., US export controls, product launches, supply chain disruptions) and their impact to corresponding share prices serving as the ground-truth.

Proposed System Design

We propose a multi-agent system composed of six specialized agents. Five domain agents monitor their respective information sources, and one Portfolio Manager agent synthesizes their insights into investment decisions.

<u>The Agents:</u> (1) Politics & News Agent monitors geopolitical events, trade policy, and government regulations via news website APIs and policy databases, tracking US-China relations, export controls, CHIPS Act developments, and Taiwan geopolitical risk. (2) Stock Market Agent monitors equity market movements, company-specific news and events, sector rotations, and trading volumes via Yahoo Finance and market data APIs, identifying unusual price movements and correlating them with potential catalysts. (3) Commodities Agent tracks raw material prices, supply chain indicators, and logistics data, monitoring

silicon wafer prices, rare earth materials, neon gas supply, and shipping costs. (4) Tech Publications Agent follows technical developments from industry conferences (IEDM proceedings), trade publications, and technical papers, tracking process node advances, EUV developments, chiplet architectures, and AI accelerator innovations. (5) Macro Agent monitors monetary policy, interest rate changes, and credit conditions via FRED (Federal Reserve Economic Data) and central bank announcements. (6) The Portfolio Manager Agent is the synthesis and decision-making agent updating the DCF model.

System Workflow: The system operates in four phases. Phase 1 (Event Detection): Each domain agent continuously monitors its information sources and flags significant events to a shared event board with source citations and relevance scores. Phase 2 (Causal Graph Construction): When a high-relevance event is detected, the Portfolio Manager initiates a collaborative session with the relevant domain agents to build a causal graph connecting the event to downstream effects on company fundamentals. Phase 3 (Multi-Agent Debate and Quantification): The Portfolio Manager facilitates a debate among the domain agents to quantify the magnitude and probability of each causal link, producing specific numerical adjustments to financial model inputs for each affected company (e.g., "NVIDIA revenue growth: -8% vs. consensus; gross margin: -100 bps"). Phase 4 (Valuation and Portfolio Construction): The Portfolio Manager feeds the consensus adjustments into DCF models to generate intrinsic value estimates, compares these to current market prices, and constructs a long-only portfolio by buying stocks where intrinsic value significantly exceeds market price (e.g., >15% upside). Portfolio performance is tracked and benchmarked against the S&P 500.

Evaluation Plan and Benchmarks

Our evaluation is grounded in a manually constructed training set of 5-10 major historical events that impacted the semiconductor value chain over the past 2-3 years. For each event, we create a ground-truth causal graph and document the actual financial impacts on the five companies. We assess the council's performance on a held-out test set across three metric categories: (1) Reasoning Quality measures event detection accuracy, causal graph overlap with ground truth (>80% target), and source grounding rate (>80% target); (2) Financial Accuracy measures MAE of predicted model input adjustments versus actual reported results and directional accuracy; (3) Investment Performance measures portfolio returns versus S&P 500, Sharpe ratio, and win rate (percentage of stocks that outperform).

Baselines: We compare the council against two key baselines: (1) Single-Agent Baseline, where a single generalist agent performs the entire reasoning and valuation process, testing whether domain specialization improves performance; and (2) the S&P 500 benchmark.

Failure Mode Analysis: We will systematically categorize errors, including detection failures (missed events), graph construction errors (incorrect causal links), quantification errors (wrong magnitude), synthesis failures (Portfolio Manager unable to reconcile conflicting views), and valuation errors (incorrect model adjustments or poor buy/sell decisions).

Group Formulation

Sameer brings expertise in software engineering from his background at Meta as well as business experience from his time at the GSB. Sam brings domain expertise in semiconductors and research with a background in the EE Phd program at Stanford. Matt contributes stock market investing experience from working as an analyst at Viking Global, a hedge fund, and access to proprietary business data sources via GSB school account.

References

[1] E. Chang, "AGI Book Volume #1, The Missing Layer of AGI: From Pattern Alchemy to Coordination Physics."
[2] G. Du, et al., "Improving Factuality and Reasoning in Language Models through Multi-Agent Debate," 2023.